Early enthusiasts

While the term “citizen science” is relatively new, dating from the 1980s, the concept and practice of using the public to observe the natural world and record data has a long pedigree. In the 1870s, small groups of ornithologists in Germany and Scotland began to collect reports on the fall migration of birds, the Scottish enthusiasts using lighthouses around the coast as observation posts. Then, in the early 1880s, the idea of collective observation was extended onto a national scale by American ornithologist Wells Cooke, who began a project to show arrival dates for migratory North American birds and provide evidence for migration pathways. Cooke’s project ran until World War II, gathering 6 million data cards on more than 800 bird species and utilizing 3,000 volunteers at its peak. In 2009, the North American Bird Phenology Program began to digitize the data from the cards, which has provided valuable evidence of changed bird migration dates and routes resulting from global climate change.

The world’s longest-running citizen science survey is the Christmas Bird Count (CBC), held each year in the US. Christmas “side hunts” of birds were a popular pastime in many rural districts of the US in the 19th century, regardless of whether the birds were suitable for eating. In 1900, Frank Chapman, an officer of the Audubon Society—named after American ornithologist and painter John James Audubon—proposed counting birds, rather than shooting them. He encouraged 27 birdwatchers to participate in the first event, and the counts then grew every year. In 2023–24, 83,186 observers submitted counts from 2,677 different locations in North and Latin America, the Pacific, and the Caribbean. The data on the distribution and number of birds has provided a huge data set for ecologists, allowing comparison over time and between habitats.